

JEE MAIN REVISION MASTER: FLUID MECHANICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Continuity Equation & Bernoulli's Principle	5 PYQs	Fluid Dynamics
Moderate	Hydrostatics, Pascal's Principle & Archimedes' Buoyancy	4 PYQs	Fluid Statics
Moderate	Viscosity, Stokes' Law & Surface Tension	3 PYQs	Surface & Transport Phenomena

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Continuity Equation & Bernoulli's Principle	$A_1 v_1 = A_2 v_2$, $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$, $v_{\text{efflux}} = \sqrt{2gh}$
Hydrostatics, Pascal's Principle & Archimedes' Buoyancy	$P = P_0 + \rho g h$, $\frac{F_1}{A_1} = \frac{F_2}{A_2}$, $F_B = \rho_L V_{\text{sub}} g$
Viscosity, Stokes' Law & Surface Tension	$v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$, $\Delta P_{\text{drop}} = \frac{4T}{R}$, $\Delta P_{\text{bubble}} = \frac{8T}{R}$, $h = \frac{2T \cos \theta}{\rho g r}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Continuity Equation & Bernoulli's Principle (Moderate Yield — 5 PYQs)

Core Principle: Conservation of mass for incompressible fluid: $A_1 v_1 = A_2 v_2$. Bernoulli's equation $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$. Torricelli's speed of efflux $v = \sqrt{2gh}$. Carburetor / atomizer suction operates on Bernoulli's principle.

■ **Caution / JEE Pitfall:** Where fluid velocity is highest (narrow section), static pressure is lowest, NOT highest!

■ **Speed Trick:** In horizontal pipes, $\Delta P = \frac{1}{2}\rho (v_2^2 - v_1^2)$. If radius is halved, area is 1/4, velocity is 4x.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Instrument fitted in the carburetor of the automobile to provide the correct mixture of air and fuel necessary for combustion works on..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A horizontal pipe carries water in a streamlined flow. At a point along the pipe, where the cross-sectional area is 10cm^2 , the velocity..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "Water flows through a horizontal pipe of varying cross-section at a rate of $0.314\text{m}^3\text{s}^{-1}$. The velocity of water at a point where the ..." [Ans: N/A]

3.2 Hydrostatics, Pascal's Principle & Archimedes' Buoyancy (Moderate Yield — 4 PYQs)

Core Principle: Fluid pressure variation with depth $P = P_0 + \rho g h$. Pascal's law of pressure transmission in hydraulic machines: $(F_1 / A_1) = (F_2 / A_2)$. Archimedes' principle: buoyant upthrust $F_B = \rho_{\text{(fluid)}} V_{\text{(displaced)}} g$.

■■■ **Caution / JEE Pitfall:** Total absolute pressure is $P = P_0 + \rho g h$; gauge pressure is only $\rho g h$. Watch whether the question asks for gauge or absolute pressure.

■ **Speed Trick:** Fraction of volume submerged for a floating body: $\frac{V_{\text{(sub)}}}{V_{\text{(total)}}} = \frac{\rho_{\text{(body)}}}{\rho_{\text{(liquid)}}}$.

Representative Source PYQs:

- [2020 Shift] Q#4 (MCQ): "Iceberg floats in water with part of it submerged. What is the fraction of the volume of iceberg submerged, if the density of ice is ρ_{ice} ..." [Ans: N/A]
- [2019 Shift] Q#2 (MCQ): "An aluminium sphere is dipped into water. Which of the following is true?..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The pressure at the bottom of a liquid tank is not proportional to the..." [Ans: N/A]

3.3 Viscosity, Stokes' Law & Surface Tension (Moderate Yield — 3 PYQs)

Core Principle: Stokes' drag $F = 6\pi \eta r v$. Terminal velocity $v_t = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta}$ (propto r^2). Excess pressure $\Delta P = \frac{2T}{R}$ (liquid drop), $\Delta P = \frac{4T}{R}$ (soap bubble). Capillary rise $h = \frac{2T \cos \theta}{\rho g r}$.

■■■ **Caution / JEE Pitfall:** 1. Terminal velocity depends on r^2 , NOT r ! If radius is doubled, v_t quadruples. 2. A bubble inside a liquid has only 1 surface ($2T/R$), while a soap bubble in air has 2 surfaces ($4T/R$).

■ **Speed Trick:** In an artificial satellite or free-fall weightlessness ($g=0$), capillary water rises to the very top of the tube, no matter how long the tube is ($h \rightarrow L$).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in ..." [Ans: C]
- [2025 Shift] Q#4 (MCQ): "While determining the coefficient of viscosity of the given liquid, a spherical steel ball sinks by a distance $h=0.9\text{m}$. The radius of the ..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "Two capillary tubes P and Q are dipped vertically in water. The height of water level in capillary tube P is $(2/3)$ of the height in Q. The ratio of the radii of P and Q is ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in a freely falling water in an elevator, then the length of the water column becomes

- (A) 10 cm
- (B) 20 cm
- (C) 30 cm [Correct]
- (D) Zero

Correct Answer: C

Solution Analysis: In normal conditions, water rises in a capillary tube because the upward force due to surface tension balances the weight of the water column. The capillary rise is $h = \frac{2T \cos \theta}{\rho g r}$. Here, g is the acceleration due to gravity. In a freely falling elevator, the effective acceleration due to gravity becomes zero: $g_{\text{(eff)}} = 0$. So the water column has no effective weight. Since there is no downward weight to balance the upward pull due to surface tension, the water will rise as much as possible in the tube. But the tube is only 30 cm long. So the water can fill the whole tube.

Benchmark Problem 2: KCET 2026 (Q#2)

Instrument fitted in the carburetor of the automobile to provide the correct mixture of air and fuel necessary for combustion works on

- (A) Pascal's Law
- (B) Bernoulli's Principle
- (C) Newton's Law of Cooling
- (D) Archimede's Principle

Correct Answer: None

Solution Analysis: The correct option is: Option B: Bernoulli's Principle A carburetor mixes air and fuel in the correct ratio for combustion in an automobile engine. Inside the carburetor, air passes through a narrow region called a venturi. When air flows through this narrow part, its speed increases. According to Bernoulli's Principle, when the speed of a fluid increases, its pressure decreases. So, the pressure in the narrow region becomes less than the pressure in the fuel chamber. Due to this pressure difference, fuel is drawn into the moving air stream. This forms the air-fuel mixture needed for combustion.

Benchmark Problem 3: KCET 2020 (Q#4)

Iceberg floats in water with part of it submerged. What is the fraction of the volume of iceberg submerged, if the density of ice is $\rho_{(i)} = 0.917 \text{ g cm}^{-3}$?

- (A) 0.917
- (B) 1
- (C) 0.458
- (D) 0

Correct Answer: None

Solution Analysis: Density of ice, $\rho_{(i)} = 0.917 \text{ g cm}^{-3}$ Fraction of the volume of iceberg submerged in water is given as $\left(\frac{\text{Volume of iceberg submerged in water}}{\text{Volume of iceberg}} \right) = \left(\frac{\text{density of ice}}{\text{density of water}} \right) = \frac{0.917 \text{ g cm}^{-3}}{1 \text{ g cm}^{-3}} = 0.917$